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Let's learn about **satellites**!

Space mission and **satellite** design course

Attitude Determination and Control System

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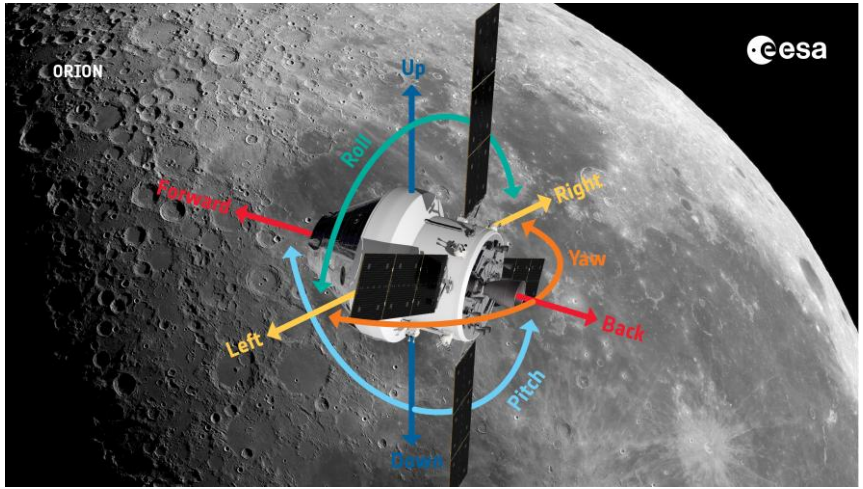
Why Attitude Matters

The orbit is not enough a satellite must also point in the **right direction**!

If attitude is wrong:
- Images are useless (camera not pointing to target)
- Downlink may fail (antenna not pointing to ground)
- Power drops (solar panels not facing the Sun)

Attitude = mission performance.



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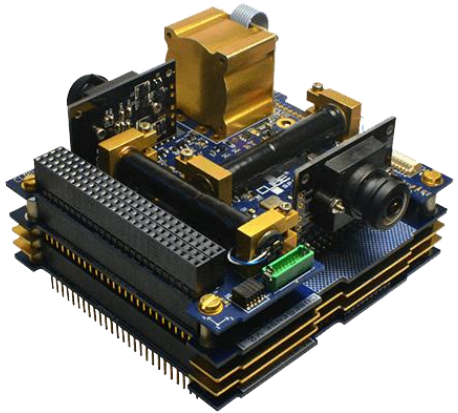
What is ADCS?

ADCS is the subsystem that:
- Understands where the satellite is pointing (determination)
- Changes and maintains the pointing (control)

Two building blocks:
- **Sensors** → measure orientation and rotation
- **Actuators** → apply torques to rotate the satellite



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Attitude Determination

Goal: estimate the satellite orientation in space.

We want to know:
- Where the satellite is pointing
- If it is rotating
- How fast it is rotating

This is done by combining measurements from **multiple sensors**.

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Main Attitude Sensors

Typical sensors used in small satellites:

Sun sensors
- Simple and low cost
- Good for coarse pointing
- Not usable in eclipse

Magnetometer
- Measures Earth magnetic field direction
- Useful for coarse attitude and for control with magnetorquers

Gyroscopes/IMU
- Measure rotation rates
- Very useful short-term
- Must be corrected over time using other sensors

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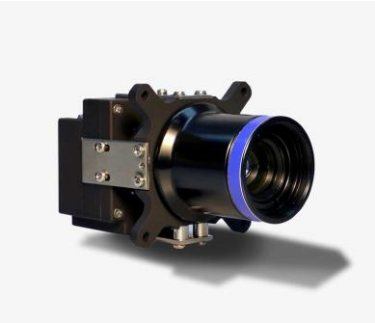
Main Attitude Sensors



Star tracker
- High-precision attitude sensor
- Higher cost, complexity, and integration effort

Earth / horizon sensor
- Useful for Earth-pointing (nadir pointing)
- Common in Earth observation missions

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Why Sensor Fusion?

Because no single sensor is perfect!

Typical limitations:
- Some sensors do not work in eclipse
- Some are noisy or low accuracy
- Some drift over time

So ADCS uses sensor fusion:
- Combine multiple sensors
- Improve accuracy and robustness

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Attitude Control

Once we know the attitude, we must control it.

Control means:
- **Rotate** the satellite to the desired pointing
- **Reject** disturbances during the mission
- **Stabilize** rotation after separation (detumbling)

Control is achieved by applying torque with **actuators**.

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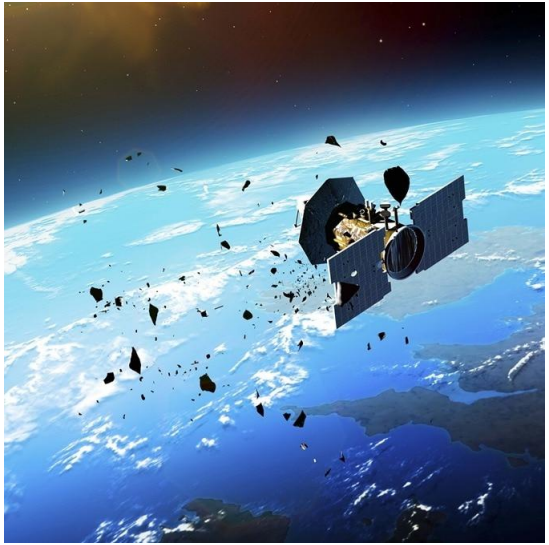
Why the Satellite Rotates in Orbit



Even if we do nothing, the satellite initial motion is disturbed.

Main disturbance sources:
- Gravity gradient (depends on shape and orbit)
- Solar radiation pressure (sunlight “push”)
- Earth magnetic field interaction (residual magnetism)
- Aerodynamic drag (for low orbits)

ADCS is employed to counteract these effects.

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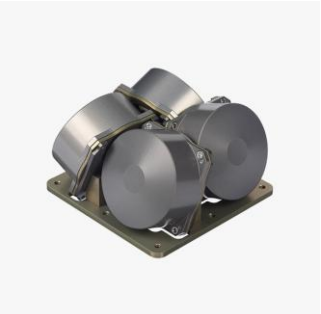
Main Actuators (Overview)

Magnetorquers
- Coils that interact with Earth magnetic field
- Very common in CubeSats
- Great for detumbling and wheel desaturation
- Limited control authority (depends on orbit and latitude)

Reaction wheels
- Spinning wheels that create control torques
- Enable accurate 3-axis pointing
- Need magnetorquers (or thrusters) for desaturation



Thrusters (when available)
- Can generate torques and also change orbit
- Higher complexity and system impact (propellant, safety, operations)

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Typical CubeSat ADCS Architectures

Basic mission (coarse pointing)
- Magnetometer + sun sensors + magnetorquers
- Suitable for low-cost missions and detumbling

High-performance mission (precise pointing)
- Star tracker + gyros + reaction wheels
- Magnetorquers used for desaturation

Choice depends on:
- Required pointing accuracy
- Payload type (optical, comms, science)
- Power and volume available
- Mission risk and budget

ADCS is a highly complex subsystem, difficult to test and, as a result, it is typically one of the most expensive components of a satellite platform!

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